

Parthiv Gajula

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Education

Wayne State University — Detroit, MI

Aug 2022 - May 2026

B.S. in Computer Science, Minor in Artificial Intelligence

Relevant Coursework:

Deep Learning, Training Large Language Models (LLMs), Intelligent Systems, Operating Systems, Database Management Systems (DBMS), Unix System Management

Technical Skills

Languages: Java, Python, C, C++, JavaScript, Ruby, R, Kotlin, SQL

Frameworks: React, [Next.js](https://nextjs.org/), Flask, Ruby on Rails, MERN, LAMP, PyTorch

Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Firebase

DevOps/Cloud: Docker, Kubernetes, Terraform, Jenkins, Nginx, Render

Testing: Jest, Selenium, E2E, ESLint

Tools: Postman, Expo, Apache, SSMS

Authentication & Security: OAuth 2.0, JWT, OpenSSL, AES, OpenVAS

AI/ML: Transformer Architecture, Self-Attention, Masked Language Modeling, LSTM Networks, Retrieval-Augmented Generation (RAG), ColBERT Retrieval, Vector Embeddings, Semantic Search, Document Chunking

Work Experience

365 Retail — Database Intern

May 2025 - Aug 2025

- Optimized SQL queries, improving execution speed by up to 45% using indexing strategies and execution plan analysis
- Managed remote database servers and configured replication between instances with filtering rules
- Tuned performance using analysis of ad-hoc queries, prepared statements, plan cache, and buffer pool behavior
- Developed custom SQL triggers for automated data processing and integrity enforcement

Tech: MySQL, SQL Server, SSMS, Replication, Query Optimization

Projects

SpeechMatch - Capstone Team Lead

Jan 2026 - Present

- Leading a capstone team developing a mobile pronunciation training platform that evaluates spoken voice against reference voice
- Coordinating directly with the project client to gather requirements, refine feature specifications, and align development milestones with stakeholder expectations
- Designed backend APIs for audio processing, OCR-based passage ingestion, and attempt evaluation

Tech Stack: FastAPI, React Native (Expo), PostgreSQL, OCR, MFA, Voice Detection Models, Docker

AWS S3 Encrypted FUSE Filesystem

Sep 2025 - Dec 2025

- Implemented a FUSE3-based filesystem that transparently encrypts files before upload to an S3-compatible bucket
- Intercepted the FUSE file descriptor cache (fdcache) to insert write operations prior to upload.
- Integrated AES-256 encryption via OpenSSL to secure data at rest before remote storage
- Implemented a SHA-256 hashing pipeline using OpenSSL primitives to maintain backward compatibility and ensure integrity
- Created a remote directory abstraction for controlled testing of encrypted file uploads

Tech Stack: AWS S3, C, FUSE3, OpenSSL, Linux, OPENSSL, AES-SHA-256

Spotify Background Recommendation App

Oct 2025 - Jan 2026

- Built an ML-driven music recommendation system that analyzes listening history to generate personalized playlists
- Applied clustering and similarity metrics to model evolving user taste profiles
- Integrated Spotify Web API for real-time track metadata and playback analytics
- Implemented hotkey ML triggers to regenerate playlists and produce monthly listening analytics reports

Tech Stack: Python, Spotify Web API, Flask, K-Means Clustering, OAuth 2.0

Team Lead @ Dearborn Hacks: Won Best Use for ElevenLabs

Oct 2025

- Team Lead for DriveMate, an AI driving assistant providing real-time conversational support for drivers
- Integrated Fetch.ai ASI-1 and built a browser-based speech-to-text / TTS pipeline using Web Speech API and ElevenLabs
- Implemented a multi-agent architecture using Python uAgents for routing, memory, and response generation

Tech Stack: Eleven Labs TTS, Flask, Web Speech API, Fetch.ai ASI-1, uAgents

Retrieval Augmented Generation(RAG) Research Study

Jan 2025 - May 2025

- Conducted a technical study of retrieval-augmented generation architectures including **REALM, DPR, ColBERT, and SELF-RAG**
- Implemented document chunking, indexing, and embedding strategies for semantic retrieval
- Evaluated **LLM-only vs RAG-based generation** using benchmark datasets such as **MS MARCO**
- Analyzed retrieval ranking behavior and limitations of token-level similarity matching

Skills: LLMs, Vector Embeddings

Ubuntu System Administration Project

Sep 2024 - Dec 2024

- Configured an Ubuntu server with SSL/TLS and HTTPS on a non-standard port (444)
- Generated and deployed custom SSL certificates using OpenSSL
- Applied security hardening to Apache2 services

Tech Stack: Ubuntu Linux, Apache2, OpenSSL, OpenVAS, Kali Linux, Bash

GrizzHacks — Won Interactive Media

Feb 2024

- Developed a full-stack music recommendation platform based on heart-rate data
- Improved BPM prediction accuracy by **80% using K-Means clustering**
- Extracted and processed audio features using librosa and integrated Fitbit biometric data

Tech Stack: PyTorch, Librosa, Fitbit API, Flask, K-Means Clustering